

National University of Computer and Emerging Sciences, Lahore Campus



Course: Introduction to Information & Communication technology
 Program: BS (Software Engineering)
 Duration: 20 Minutes
 Paper Date: 10-September-2025
 Section: BSE-1A
 Exam: Quiz 1

Course Code: CL-1000
 Semester: Fall 2025
 Total Marks: 10
 Weight: 2
 Page(s): 2
 CLO: 2

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Answer the questions in the given available space.

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Q:1 The student ID is given as $(5678)_{10}$. Represent it in:

[5 Marks]

1. Binary
2. Octal
3. Hexadecimal

1- Decimal to Binary
 Taking LCM by 2

2	5678
2	2839-0
2	1419-1
2	79-1
2	39-1
2	19-1
2	9-1
2	4-1
2	2-0
1	1-0

$(5678)_{10}$
 $=$
 $(100111110)_2$

X

2- Decimal to Octal
 Taking LCM by 8

8	5678
8	79-6
8	19-7
	1-1

$(5678)_{10} = (\cancel{1436})_8$
 $(1176)_8$

X

3- Decimal to hexadecimal
 Taking LCM by 16

16	5678
16	354-14 → E
16	22-2
	1-6

$(5678)_{10} = (162E)$

✓

Q:2 The number $(10101011)_2$ is stored in computer memory. If the system uses signed magnitude, what is its decimal value? [5 Marks]

$$\begin{array}{cccccccc} 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\ (1 & 0 & 1 & 0 & 1 & 0 & 1 & 1) \end{array} \begin{array}{l} 2 \\ 2 \\ 2 \\ 2 \\ 2 \\ 2 \\ 2 \\ 2 \end{array} = (1 \times 2^7) + (0 \times 2^6) + (1 \times 2^5) + (0 \times 2^4) + (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0)$$

$$= 128 + 32 + 8 + 2 + 1$$

$$= 171$$

The decimal value of $(10101011)_2$ in decimal will be $(171)_{10}$

2

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